

Jinqiang Yu

PH.D | RESEARCH SCIENTIST | OMNI / MULTIMODAL LLM PRE-TRAINING & MID-TRAINING

Sydney, Australia

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Summary

Research Scientist specializing in **Omni / multimodal foundation models**, with hands-on ownership of **Audio-Visual LLM (AVLM) pre-training and mid-training** across data, model adaptation, training, evaluation, and serving efficiency. Built multimodal foundation models for large-scale production scenarios, with research spanning **multimodal data curation, continual / domain pre-training, model adaptation, token compression, MoE / dense scaling, and AutoResearch**. PhD in AI with publications in Neurips, AAAI and other top venues.

Experience

TikTok

Sydney, Australia

Research Scientist

Jul 2024 - Present

- Lead **Audio-Visual LLM (AVLM) pre-training and mid-training**, owning the end-to-end (manual & auto)-research loop from data construction and training recipes to multimodal evaluation, root-cause analysis, and production validation.
- Developed **AVLM 0.8B/2B/4B/9B/27B/35B-A3B** model families on multiple VLM backbones, **preserving vision** capability while adding **strong audio and audio-visual understanding capabilities**, achieving outstanding performance for moderation and other downstream tasks.
- Drove model-efficiency research including **4x audio-token compression** with negligible quality regression, reducing incremental GPU demand from audio serving from approximately **1.30x** to **1.15x** versus the VLM backbone.
- Built **domain continual mid-training pipelines over large-scale corpora**, covering data collection, quality filtering, deduplication, mixture design, and systematic ablation across model scales and architectures. Leading significant improvement in downstream.

Monash University / OPTIMA

Melbourne, Australia

PhD Researcher

Apr 2021 - Jul 2024

- Conducted research in explainable and interpretable machine learning, including explanations, feature attribution, interpretable models, and applications to NLP, CV, and LLMs.
- Extended explainability research to **LLM adaptation and reasoning**, including prompt engineering, parameter-efficient finetuning, and reinforcement-learning-based methods.

Education

Monash University

Melbourne, Australia

PhD in Artificial Intelligence

Feb 2021 - Jul 2024

- **Thesis:** Explainable AI for Machine Learning – formal explanations, interpretable models, feature attribution, and applications of explainability to modern ML/LLM systems.

Selected Projects

Audio-Visual Foundation Models: Pre-training, Mid-training & AutoResearch

Sydney, Australia

TikTok

Oct 2025 - Present

- Built the **full Audio-Visual LLM (AVLM) research stack**: multimodal data collection/cleaning, audio and vision encoders/projectors, dense & MoE VLM → AVLM initialization, data construction, staged training mixtures, modality-balanced evaluation, and downstream feedback loops.
- Established **hundreds-billion tokens of mid-training pool** and data engine; validated up to **+7.5 pp** downstream gains from data selection, filtering, cleaning, and labeling strategies.
- Developed **0.8/2B/4B/9B/27B/35B-A3B AVLM models** whose in-domain audio, vision, and Audio-Vision (AV) performance surpassed SOTA similar-size omni models; in particular, 27B & 35B-A3B models outperform **Gemini-2.5-Pro** overall.
- Achieved **4x audio-token compression** with negligible quality loss, reducing the incremental GPU requirement of audio conditioning from 1.30x to 1.15x while preserving general and downstream capability.
- Productized VLM → AVLM adaptation in **AutoResearch** that automate model assembly, training/evaluation recipes, diagnosis, and iteration reducing manual effort to **0.2x**.

Large-Scale LLM Continual Pre-Training for Domain Foundation Models

Sydney, Australia

TikTok

Jan 2024 - Sep 2024

- Led **data, training, model, and evaluation** research on continued-pre-training recipes at **572B-token scale** across dense and MoE backbones from **1.2B to 32B**.
- Built an end-to-end pipeline for mixture curation, deduplication, quality filtering, and domain balancing, sustaining approximately **290B tokens/week** throughput.
- Improved validation construction and downstream correlation for rapid model/data selection; systematic scaling and mixture ablations produced an average **8% improvement** across downstream tasks.

Explainable and Interpretable Machine Learning

Melbourne, Australia

TikTok

Feb 2021 - Jul 2024

- Designed exact and approximate methods for ML models, formal feature attribution, and succinct abductive / contrastive explanations; applied them to NLP, CV, healthcare, and software areas.
- Investigated explanation-guided **LLM adaptation** using prompt engineering, parameter-efficient fine-tuning, and reinforcement learning.

Selected Publications

From Failure Taxonomy to Intervention: A Diagnostic Methodology for Industry-Scale AVLM in Video and Live-Streaming Platform Moderation

Shuchang Ye, Jinqiang Yu, al.

Neurips, 2026

A formal explainer for just-in-time defect predictions

Jinqiang Yu, Michael Fu, Alexey Ignatiev, Chakkrit Tantithamthavorn, Peter Stuckey

TOSEM, 2024

From Formal Boosted Tree Explanations to Interpretable Rule Sets

Jinqiang Yu, Alexey Ignatiev, Peter J. Stuckey

CP, 2023

Eliminating the Impossible, Whatever Remains Must Be True: On Extracting and Applying Background Knowledge in the Context of Formal Explanations

Jinqiang Yu, Alexey Ignatiev, Peter J. Stuckey, Nina Narodytska, Joao Marques-Silva

AAAI, 2023

Technical Skills

Foundation Models	Omni-LLM / Multimodal LLM , Audio-Visual LLM, LLM/VLM, dense & MoE architectures, Transformers
Pre/Mid-Training	Continual/domain pre-training, multimodal mixture design, data curation, quality filtering, deduplication, curriculum/recipe design, scaling studies, model adaptation
Post-Training	Supervised fine-tuning, PEFT, RL/RLHF, prompt engineering, reasoning, AI labeling
Evaluation & Efficiency	Multimodal benchmarking, ASR/perception/reasoning evaluation, failure taxonomy/RCA, proxy evaluation, token compression, inference/serving optimization, AutoResearch
Programming & Frameworks	Python, PyTorch, Hugging Face, NumPy, Pandas, C++, Java, R, MATLAB
Systems & Data	Linux, Git, Docker, Slurm, AWS, Spark, SQL, MongoDB

Scientific Activities & Awards

2024-2026	PC member of the AAAI Conference on Artificial Intelligence
2024	PC member of the International Joint Conferences on Artificial Intelligence (IJCAI-2024)
2020	Best Paper Award: My paper “Computing Optimal Decision Sets with SAT” has been selected for the Best Paper Award for the CP/ML Track of CP 2020.